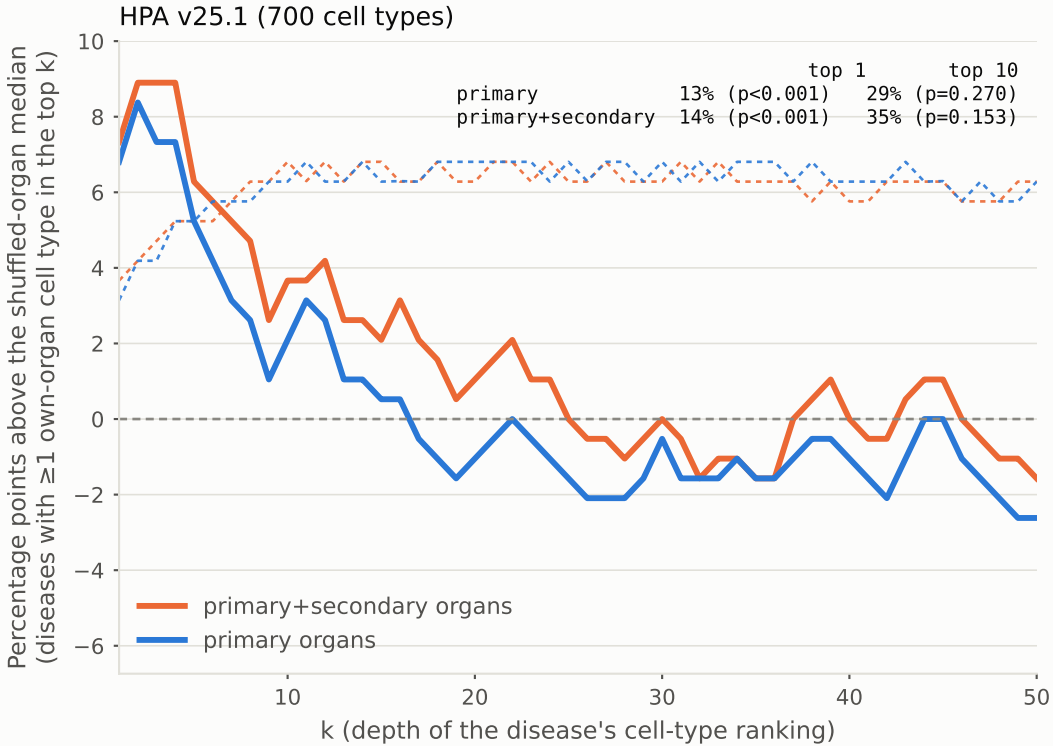
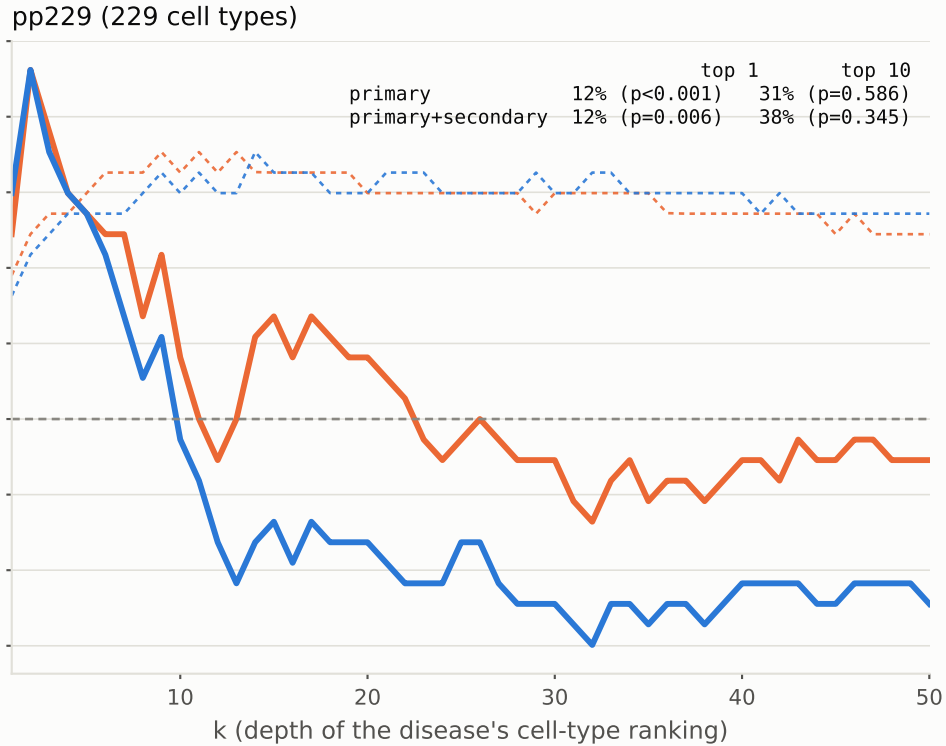
Does adding secondary organs help? Excess hit rate over the shuffled-organ baseline

Lines: observed minus the median of its own shuffled-organ null (each definition has its own null, because a wider organ set raises the chance of a hit).
Dashed: upper 95% limit of that null (same colour). A line above its dashed limit = more own-organ hits than chance.
Boxes: share of diseases with ≥1 own-organ cell type in the top k, and permutation p.





Secondary organs on their own (primary-organ cell types excluded; only diseases that have secondary organs), observed vs shuffled:
hpa700 (68 diseases): top 1 3% vs 3% shuffled, top 10 16% vs 16%; pp229 (61 diseases): top 1 2% vs 7% shuffled, top 10 26% vs 28% → no excess over chance.